

Assessing Racial Injustices and Income Gaps in Chicago through Water Utility Debt

1. Introduction

Access to clean water is a fundamental human right and a basic biological necessity, making it a non-discretionary resource within residential households. The demand for water is highly inelastic, meaning that as the price of water bills increase, households cannot substantially reduce consumption (Luby, Polasky, and Swackhamer 2018). Consequently, water utility functions as an essential household expense alongside rent, electricity, and gas—it is a resource that cannot be substituted. These bills function as recurring monthly obligations that must be paid regardless of whether an individual owns or rents their home.

As cost of living increases, essential bills have also faced inflation costs. Water and wastewater costs in the United States have been rising three times faster than inflation, causing water utility bills to skyrocket. This effect is largely driven by a crucial and widespread need for improving and updating infrastructure issues in outdated sewer and water systems that are increasingly noncompliant with environmental regulations (Cardoso and Wichman, 2022). Therefore, when increases in utility bills occur, residents—particularly in low-income households—who have little to no control over public water systems must continue to make financially straining payments or risk accumulating debt. When individuals and households accumulate debt for essential household bills such as water utility, this can serve as an indicator of a broader financial hardship and a much more deeply rooted economic inequality issue (Ponder and Omstedt, 2019).

In many American metropolitan areas, economic disparities and wage gaps are vividly apparent. Chicago in particular is characterized by a history of highly disproportionate wages and instances of redlining. Presently, there is still substantial socioeconomic variation across ZIP codes, reflecting a long-standing pattern of segregation and uneven economic development. (Schwegman, 2025). Consequently, Chicago provides the opportunity to investigate a compelling case study on understanding income inequality and demographic composition through the lens of water utility debt. Existing research reveals that in Chicago there are large clusters of neighborhoods that face issues of water unaffordability, which can be a sign of low income communities. This indicates that water utility debt is a factor of uneven distribution of wealth at a local level (Cardoso and Wichman, 2022).

Due to such stark economic differences between neighborhoods, this paper aims to show that examining water utility debt at the ZIP code level can serve as a proxy for understanding the spatial economic differences in urban areas. In cities with a long history of racial segregation, patterns of utility debt can also reflect racial inequality, given the longstanding association

between low income communities and racially segregated neighborhoods. To better understand these relationships and utility debt patterns, this paper will therefore investigate the relationship between racial demographics and aggregate utility debt on the ZIP code level while controlling for income levels to address the question of whether water utility debt in Chicago varies systematically across neighborhoods and whether these patterns are associated with racial demographics after accounting for income differences.

2. Context

Rising Infrastructure Costs and Residential Burdens

The financing of United States utilities is caught in a cyclical loop. Many utilities rely heavily on payments from their own customers to finance operations, management, and infrastructure projects (Patterson and Doyle, 2021). Subsequently, high utility bills are frequently driven by outdated infrastructure that require drastic upgrades to meet environmental standards (Cardoso and Wichman, 2022). As infrastructure costs increase, utilities are forced to raise their rates to generate revenue to finance these repairs and upgrades. This becomes an even larger challenge for utilities as federal funding for such projects decreases. By 2017, state and local governments were responsible for 96% of their own water utility financing, causing local utilities to generate revenue primarily through customer rates and local taxes. These in turn placed the financial burdens onto residents, especially impacting low-income households that struggle to afford the rising cost. In turn, many low-income communities are limited in their ability to invest in local infrastructure improvements and upgrade their equipment leading to poor quality utility systems reinforcing the financing cycle (Patterson and Doyle, 2021).

Utilities that serve a large proportion of low income residents that face higher levels of financial hardships often face greater difficulty generating revenue through customer payments. At the same time, these communities are more vulnerable to rising utility costs because income growth for low-income households have not kept pace with increasing water service expenses. Over time, an income gap has emerged in which higher income households are slowly increasing their income while low-income households remain stagnant or at a slower pace, resulting in a struggle to keep up with rising living costs and inflation. As a result, low-income households use a larger share of their income to pay their water utility bill thus widening the income gap. Research shows that a minimum wage earner spends 10 hours of labor each month to afford basic water services (Teodoro and Saywitz, 2020). Low income households spend about 16.5% of their disposable income, their salary after paying for essential services and goods such as rent, food, and energy, on water services.

An affordability gap is characterized as the difference between the costs of providing water services and the ability to pay for those services. As a non discretionary item that is used for drinking, sanitation, hygiene, and living, the inability to afford water services has a larger existential implication than debt. This can, in the long term, create a public health crisis as utilities struggle to pay for infrastructure costs through residential payments due to local utilities dependencies on local community investment (Patterson and Doyle, 2021).

Water Shutoffs and Public Health Implications

In the United States, where there is a lack of a federal policy or national statute guaranteeing access to clean water; the implications of water unaffordability go far beyond consumer debt and utility financing structures. The consequences of utility debt can lead to water shut offs, which create additional disadvantages and resource allocation challenges for low-income communities. In urban cities, research shows that regions with larger shares of racial and ethnic minorities are more likely to experience issues with water affordability that lead to a higher chance of water shut offs. However, this is only partially explained by utility debt, suggesting broader structural inequalities in income, infrastructure investment, and resource allocation also shape access to water services. (Sarango, Kim, and Harlan, 2025)

The price increase of water utilities is unique compared to other utilities such as gas and electricity, as the cost of water and sewage services have risen faster than those of other utility services (Cooley, Shimabuku and DeMyers, 2022). Oftentimes, water shut offs are a method used by utilities to leverage payment from households who are not paying their bills and to maintain the financial stability of water systems and services (Walton, 2018). A study that examined nine urban cities' water shut offs showed that of communities with the highest portion of people of color, 6.26% of respondents experienced water shut offs compared to only 1.03% in majority white communities (Sarango, Kim, and Harlan, 2025).

Instances of water shut offs have serious public health consequences that are predominantly affecting minority and economically disadvantaged populations. Research shows that it not only affects biological health and development it can also lead to emotional and psychosocial distress (Gaber et al. 2021). Even receiving a notice of a potential water shut off has been associated with increased stressed levels in households. Water insecurity can also lead to changes in behavior that negatively affect an individual's household routine including a reduction in routine hygiene practices such as bathing, cleaning, laundry, and even washing hands (Gaber et al. 2021). This increases the risk of infectious disease transmission (Pruss-Ustun, A., et al. 2019).

Additionally, unreliable access to drinking water and water services in general mean that households must receive water from elsewhere given the inelastic demand for water. This has

shown to have an increased reliance on bottled water. This expensive but necessary habit also serves to widen the wage gap since disadvantaged households are forced to spend their disposable income on water bottles(Gaber et al. 2021).

Spatial Inequality

While water affordability challenges exist across multiple U.S. cities, their issues often exhibit spatial patterns across neighborhoods. This makes Chicago an important and relevant case for examining how utility burdens and affordability challenges vary across urban communities. Chicago has a history of residential segregation and redlining that produced significant differences in income levels, infrastructure investment, and racial demographics throughout neighborhoods.

This phenomenon can be traced back to federal housing policies implemented during the Great Depression. In 1933, under the Roosevelt administration, the Home Owners' Loan Corporation was created for finding households with defaulted mortgages and re-issuing more affordable, low interest loans. However between 1933-1936 the department soon issued one million loans across the United States. As a result, the HOLC began to evaluate lending activity using residential security maps which categorized neighborhoods from low risk, grade A, to high risk, Grade D. These maps displayed racially discriminatory grading structures which in turn led minority neighborhoods being deemed as risky.

Although many of these neighborhoods were already economically disadvantaged prior to the creation of the HOLC, the use of these maps reinforced patterns of limited investment in communities with larger minority populations. Additionally, many residents of these neighborhoods were targeted to not receive funding from the Federal Housing Administration created in 1934 which insured private mortgages. Discriminatory policies resulted in less than 2% of the FHA insured mortgages going to Black Americans between 1945-1959 (Schwegman, 2025). These discriminatory housing policies contributed to long lasting racial segregation and uneven investment across urban neighborhoods. Historical redlining and uneven lending practices shaped spatial inequalities in cities such as Chicago. These issues lead to infrastructure inequality, economic disadvantages, and insecurity among various services across neighborhoods (Schwegman, 2025).

These historical patterns remain relevant today and are highlighted through insecurity of basic resources, income inequality, and wage gaps. Research suggests that disparities are not randomly distributed but instead tend to cluster geographically meaning that areas experiencing economic struggles are often located near other disadvantaged neighborhoods (Cardoso and Wichman, 2022). Examining water utility debt at a ZIP code level can therefore help identify

regions of Chicago where economic disparities and infrastructure issues have persisted over time. By looking at these spatial patterns, we can explore and understand how longstanding inequalities shape resource allocation and access to basic services like water and sewage.

3. Literature Review

Research on Water Affordability

What are the various ways water affordability is measured in the current literature?

Research on water affordability focuses on determining the threshold at which water services become a financial burden for households. There are multiple ways to measure this value. A common way is to create a ratio of how much a household pays for water services relative to their income. The Environmental Protection Agency (EPA) threshold for determining high burden of water utilities is when the bill exceeds 4.5% of a communities median income. However many scholars have scrutinized this methodology arguing that median income does not accurately capture the financial strain of the vulnerable populations and using a threshold of 4.5% is an arbitrary measurement.

Cardoso and Wichman (2022) measure water affordability in their paper by first categorizing water and sewage as separate components of household utility expenses. They measure affordability at a threshold of 4.5% of a household's annual income rather than using a community level median income measure. This captures affordability burdens at an individual household level rather than relying on a community based calculation. However Patterson and Doyle (2021) and Teodoro (2018) propose an alternative calculation of measuring affordability as a ratio of working hours required for an individual to pay their water utility bill. Teodoro (2018) described a rule of thumb suggesting that a four person household should not spend more than eight hours of labor at minimum wage to afford their water utility bill.

Using the framework developed by Patterson and Doyle (2021), Cardoso and Wichman (2022) calculate that one day of labor corresponds to approximately 4.6% of monthly income which closely aligns with the EPA affordability threshold. By adopting a measurement threshold similar to the EPA's, these studies are able to compare their findings to federal standards. Although an important distinction is that the EPA applies this standard at the community median income level while Cardoso and Wichman (2022) do so at a household level receiving more precise estimates of affordability burdens (Cardoso and Wichman, 2022).

Water Shutoffs and Household Impacts

What happens when a household faces or is categorized as a high burdened household?

Many researchers have studied the effects of households carrying large amounts of utility debt. A main consequence of this is water shut offs that have been extensively researched to understand the demographic of neighborhoods or communities that face the mental and health burdens of unreliable access to water. Detroit, Michigan is a popular case study due having a large low income and poverty population and a history of water insecurity issues. It serves as a prime example of understanding the consequences of water shut offs and how they burden low income populations the most.

Ponder and Omstedt (2019) examine the regions of Detroit that have a history of water shut offs and the relationship between regions with shut offs and racial demographics. They find that water shut offs are geographically clustered in regions where there is high utility debt and large demographics of Black neighborhoods. This research shows that water shut offs are not a random action done by utilities. The authors suggest that they are a strategic mechanism used to continually disadvantage already low income populations and cause them to suffer more. Several studies have examined how water shut offs are distributed across low income communities and that they disproportionately affect low income and minority populations.

Ponder and Omstedt (2019) attribute these actions of water shut offs done in specific communities as a form of racial violence in the form of infrastructure degradation. They argue that instead of comments and actions of direct racism, discrimination in urban cities is also done through financial systems through municipal debt, fines and penalties and utility fees. Because utilities are essentially financial systems, they must prioritize revenue and thus target marginalized communities through large fees and threatening water shut offs. As discussed in the previous portion, these systems, neighborhoods, and jurisdictions were created on the basis of segregation and racial targeting. Therefore, these shut offs and violations are another form of segregation and not coincidental actions.

Other studies have researched the effects of water insecurity and water shut offs and an individuals and households development, mental, and physical health. Gaber et al. (2021) examines the effects of water insecurity on households in Detroit and finds that unreliable water access is associated with psychological distress and behavioral changes such as reduced hygiene practices. The results of the paper showed statistical significance in that even just the threat of water shut offs leads to an increase in psychological distress and that this distress is related to water insecurity rather than income. This means that the distress itself is formed through unreliable water sources, not just general financial hardships. Additionally, the psychological stress of moving from having access to water to no access at all is a much more significant stressor than stagnantly not having water access. Residents that continually do not have access to water receive less distress than residents who are threatened with no access. This research also showed that participants reported having a decrease in routine hygiene which can create health risks (Gaber et al. 2021). Participants reported that they reduced activities such as bathing, laundry, and handwashing in an attempt to conserve limited water or when threatened with shut offs.

Research Gap and Contribution

Previous research on water insecurity has touched on topics such as measuring affordability and implications of water shut offs and household impacts. These studies use various research methods including in person survey analysis, regression analysis of shut offs, and spatial analysis identifying clusters of neighborhoods that are affected. However, it is less common for studies examining water insecurity and affordability to use aggregate utility debt balances as a measure of financial burden. This paper addresses this gap by examining the relationship between aggregate debt at a ZIP code level and race while controlling for income. Specifically, this study asks whether racial demographics in neighborhoods are associated with high levels of utility debt across regions in Cook County.

4. Research Design

This study examines the relationship between racial composition on a ZIP code level and aggregate past utility debt within Cook County. Specifically, the analysis further investigates whether ZIP codes with higher proportions of Black residents face larger aggregate utility debt and are faced with higher burdens that may be due to financial hardship. The analysis also evaluates if this relationship persists after accounting for different economic status across neighborhoods by controlling for income and if that makes a significant difference in the relationship between utility debt and community demographics.

Data

Obtaining and Understanding Utility Debt Data

The outcome or dependent variable of this analysis lies in the utility debt. I obtained this data from the City of Chicago's Department of Finance annual reports on outstanding past due utility debt which I secured from the years of 2022-2024. This data is organized in two tables one of which shows utility debt from each ZIP code for various service classes or types of building (Domestic residential occupancy, single family dwellings, two unit-residential structure, three unit residential structure etc). The following table, also organized by ZIP code shows the debt for each type of structured building and includes information on how many days past due for the total number of accounts that have acquired any debt over the year.

This data, although organized, was difficult to clean due to the structure of the data being in a pdf. To turn it into a csv I had to extract and clean the data to ensure everything was organized and read properly. I also created new variables to use when running my regressions. For example, an issue with ZIP code level data is that population is unaccounted for - various ZIP codes have various population sizes. To combat this, I used census population data to create

a variable called “debt per capita” which showed debt for a ZIP code based on population size rather than just total debt from each ZIP code. This way for ZIP codes that have a larger population than others, the aggregate debt calculation is scaled. Additionally, to address the distribution of debt, I used the natural log of the debt per capita.

Obtaining and Understanding Demographic Data

To understand the relationship between debt and racial demographics, I used American Community Survey (ACS) data conducted by the U.S. Census Bureau at the ZIP code level from 2019 - 2023. ACS data is calculated over the course of five years to create more stable data points. I included population, median household income, and racial composition of each ZIP code. I also used the population and race data to create the percentages of Black, Hispanic, Asian, and nonwhite variables to use for regression modeling.

Final Dataset Construction

To create the final data set used for regression modeling I merged utility and census data on ZIP codes. I also subsetting the ZIP codes to only include those within the Cook County area since there are many ZIP codes that are out of state in the water utility data set. Once I subsetting for the Cook County ZIP codes, I merged the demographic data to create the regression CSV to include variables such as total past due utility debt aggregated for each ZIP code, debt per capita, median household income, racial composition for each ZIP code, and population size.

Regression Models

To estimate the relationship between racial composition and utility debt I created three ordinary least squares (OLS) regression models.

Model 0:

This is a simple bivariate regression model to examine the relationship between debt burden and the share of black residents in each ZIP code.

$$\log(Debtpercapita)_z = \beta_0 + \beta_1 \%Black_z + \epsilon_z$$

This model evaluates whether ZIP codes with larger Black populations experience increased debt burdens.

Model 1:

This OLS regression examines the relationship between Black populations at each ZIP code level and their debt burdens while controlling for income.

$$\log(\text{Debtpercapita})_z = \beta_0 + \beta_1 \% \text{Black}_z + \beta_2 \text{Income}_z + \epsilon_z$$

Looking at this relationship allows us to understand how much of the debt burden can be explained by racial demographics when accounting for economic differences such as income across ZIP codes in Cook County.

Model 2:

$$\log(\text{Debtpercapita})_z = \beta_0 + \beta_1 \% \text{Black}_z + \beta_2 \% \text{Hispanic}_z + \beta_3 \text{Income}_z + \epsilon_z$$

This final model looks at racial composition of each ZIP code and understands the significance of different demographics in each ZIP code and its relationship to debt burden.

5. Preliminary Analysis

Regression Results

Regression Results: Utility Debt and Racial Composition

	<i>Dependent variable: Log Debt Per Capita</i>		
	Model 0 (1)	Model 1 (2)	Final Model (3)
Intercept	2.011*** (0.175)	13.820** (5.592)	14.496** (5.837)
log_income		-1.030** (0.488)	-1.096** (0.514)
pct_asian			0.684 (1.656)
pct_black	4.392*** (0.440)	3.333*** (0.665)	3.361*** (0.670)
Observations	225	225	225
R ²	0.309	0.322	0.323
Adjusted R ²	0.306	0.316	0.314
Residual Std. Error	2.030 (df=223)	2.014 (df=222)	2.018 (df=221)
F Statistic	99.583*** (df=1; 223)	52.797*** (df=2; 222)	35.123*** (df=3; 221)

Note:

* p<0.1; ** p<0.05; *** p<0.01

Regression Analysis

Looking at Model 0, we can see that the relationship between log debt per capita and percentage of Black population in each ZIP code is a highly statistically significant value. This coefficient indicates that when there is a 10% point increase in share of Black residents, there is approximately a 44% increase in debt per capita. This indicates a strong positive relationship between the share of Black residents in a ZIP code to debt burden for water utilities.

In the second model, where we controlled for income, this shows results slightly different from the previous model. The coefficient has decreased from 4.39 to 3.33. This indicates that a portion of the relationship of debt burden is partially explained by income. The coefficient for log income is -1.03 showing a negative relationship between increasing income and decreasing debt. This means that ZIP codes with higher median incomes have lower debt burdens within Cook County showing that neighborhoods with higher median incomes are less likely to accumulate water utility debt.

Finally look at the last model, the coefficient for Black demographics has increased, remaining stable and highly significant. However, the adjusted R^2 value has decreased indicating that including a variable for percentage of Asian residents in each ZIP code does not improve the model. Since this value is not statistically significant, there is no strong evidence of the relationship between debt burden and Asian population share. Comparing the adjusted R^2 values for all three models, we can conclude that this model represents or explains approximately 30% of the variation in debt per capita across ZIP codes in Cook County.

6. Conclusion

This study sought out to understand whether racial composition of neighborhoods is associated with differences in utility debt burdens across ZIP codes in Cook County. The central research question was to determine if water utility debt can serve as a proxy to look at racial inequality after controlling for income. The history of segregation and racial distribution throughout Chicago is an issue that involves many factors. Understanding difficulty accessing non-discretionary resources such as water is an important component of evaluating resource allocation within urban communities. The results of the regression models indicate that race, especially ZIP codes with high distribution of Black residents is a large indicator and serves a strong relationship with debt burden. Utility debt is not even distributed throughout neighborhoods, those with a higher proportion of Black populations face a large disadvantage that could lead to struggles in physical and mental health.

Implications

Understanding the larger implications of this study is that this logic could be and has been applied to other urban communities. Many communities such as Chicago are built on the basis of disinvestment and lack of housing and infrastructure for low income neighborhoods - often with a large proportion of not white residents. This can be used to inform policy regarding resource allocation and ensuring that residents in these regions with high debt burdens require extra assistance and the consequence of not receiving it can be physical health and increased distress. Policies that could better assist with resource allocation such as utility assistance programs, government benefits such as SNAP, and targeted outreach are examples of potential outcomes of this study. From the literature review, there have been similar studies across the United States indicating that this research is externally valid and can be applied to other urban communities.

Limitations

While the analysis provides a strong insight into the relationship between racial demographics and utility debt at a ZIP code level, there are several limitations to address. The first of which is regarding the utility debt data. Although the data set provided an extensive look at debt burdens by ZIP code, it is important to mention that if a ZIP code within Cook County did not have an account that went into debt throughout the year, there would not be any data for that making this list of Zip codes incomplete. Another issue is that oftentimes if there is no consequence for utility debt such as water shut offs, many accounts might opt out of paying their bill. To better understand the possible consequences, it would be helpful to look at the Department of Finance provided shut off data. This would show if residents who are forgoing their utility bill are experiencing consequences of debt through actions such as shut offs as other urban cities like Detroit are. Another issue is that utility debt might not always equate to an account not being able to afford their bill. Other reasons for an account having debt could reflect billing disputes, change of personal information, or administrative issues. These limitations offer potential ways to improve the external validity of this study.

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